

Your grade: 100%

Your latest: 100% • Your highest: 100% • To pass you need at least 80%. We keep your highest score.

Next item →

1. What background topics are essential to being able to derive and understand Kalman filters? (Select all that apply.)

1 / 1 point

☒ State-space dynamic systems.

 **Correct**
Yes, which is why we study them next week.

☒ Probability theory and stochastic systems.

 **Correct**
Yes, which is why we study them in week 3 of this course.

☒ Least-squares estimation theory.

 **Correct**
Yes, which is why we study this topic in the second course.

☐ Artificial intelligence.

2. What is sequential probabilistic inference?

1 / 1 point

☐ A special type of state estimator that applies only to linear systems. The present state estimate is computed recursively based on the prior estimate and new input and output measurements.

☐ A general type of state estimator, of which the Kalman filter is a special case. The present state estimate is computed afresh every timestep based on the entire history of input and output measurements.

☒ A general type of state estimator, of which the Kalman filter is a special case. The present state estimate is computed recursively based on the prior estimate and new input and output measurements.

☐ A special type of state estimator that applies only to linear systems. The present state estimate is computed afresh every timestep based on the entire history of input and output measurements.

 **Correct**
Yes, this is correct.

3. In addition to the lesson notes and videos, what other course materials are provided to enhance and assess your learning? (Select all that apply.)

1 / 1 point

☐ A free copy of a textbook on Kalman filtering.

☒ Ungraded Coursera Labs Jupyter notebooks.

 **Correct**
Yes. These will allow you to experiment with Octave code that implements different kinds of Kalman filters, and will sometimes be used to compute answers to quiz questions.

☒ Graded (summative) quizzes.

 **Correct**
Yes. There will be one of these every week.

☒ Ungraded (formative) quizzes.

 **Correct**
Yes! In fact, this quiz is one of them.